

Just for you all to understand where to use, what you are all learning!

1. K-Nearest Neighbors (KNN)

1. A hospital has records containing patients' age, blood pressure, and glucose level. It wants to classify a new patient by looking at patients with similar measurements. Which approach best describes KNN?

- A. Follow a series of learned if-else rules
- B. Find similar patients and use their labels to classify the new patient
- C. Fit one straight line through all patients
- D. Build many decision trees

2. A KNN model for identifying plant species uses `n_neighbors=5`. A new plant's five nearest neighbors include three roses and two lilies. What will a standard KNN classifier predict?

- A. Rose
- B. Lily
- C. Both species
- D. It cannot make a prediction

3. A student builds KNN using patient age (18–90) and annual medical expenditure (0–500,000). The model seems to rely mostly on expenditure. What should the student consider?

- A. Scaling the features before fitting KNN
- B. Increasing the number of target classes
- C. Removing the training labels
- D. Changing the model to regression without checking the task

4. A team wants to predict the *price* of a house using prices of similar houses nearby. Which Python model fits this task?

- A. `KNeighborsClassifier`
- B. `KNeighborsRegressor`
- C. `GaussianNB`
- D. `LogisticRegression`

5. A researcher uses `KNeighborsClassifier(n_neighbors=1)` to classify noisy medical records. Training accuracy is very high, but test accuracy is poor. What is a reasonable explanation?

- A. One unusual training example can strongly affect a prediction
- B. KNN cannot classify medical records
- C. KNN has stopped using distance
- D. The model has too many decision trees

6. A student increases `K` from 3 to 200 in a small classification dataset. The model begins predicting the majority class for many different samples. Why?

- A. A large neighborhood can hide local patterns
- B. KNN automatically changes the target labels
- C. Larger `K` always increases training accuracy
- D. Feature scaling is no longer possible

7. A wildlife team has labeled measurements from 50,000 animals. Predictions with KNN are slower than expected. What is a likely reason?

- A. Predicting may involve searching many stored examples for nearby animals
- B. KNN retrains a random forest for every animal
- C. KNN must draw a graph for every prediction
- D. KNN cannot store more than 100 examples

8. A student wants closer patients to influence a KNN prediction more than patients farther away. Which setting is most relevant?

- A. `weights="distance"`
- B. `weights="uniform"`
- C. `max_depth=5`
- D. `criterion="entropy"`

9. A team is choosing between K values of 3, 5, 7, and 9. What is an appropriate way to select K?

- A. Compare their performance using cross-validation on the training data
- B. Choose the K that gives the highest score on the final test set repeatedly
- C. Always choose the largest K
- D. Choose K based only on the number of feature names

10. A student writes `model = KNeighborsClassifier(n_neighbors=5)` and then wants to train it. Which line should come next?

- A. `model.fit(X_train, y_train)`
- B. `model.predict(y_train)`
- C. `model.score(X_train)`
- D. `model.transform(y_train)`

2. Naive Bayes

1. An email service wants to classify messages as spam or legitimate using the words in each email. Why might Naive Bayes be a useful starting point?

- A. It can use word-based evidence to compare the possible classes
- B. It requires no labeled emails
- C. It understands every sentence exactly as a human does
- D. It measures the distance between email senders

2. A student converts emails into word-count features using CountVectorizer. Which model is a common choice for these features?

- A. MultinomialNB
- B. GaussianNB
- C. KNeighborsRegressor
- D. DecisionTreeRegressor

3. A medical dataset contains continuous measurements such as temperature, heart rate, and blood pressure. The student wants to try a Naive Bayes model that assumes a Gaussian distribution for each feature within each class. Which model should they choose?

- A. GaussianNB
- B. BernoulliNB
- C. MultinomialNB
- D. RandomForestRegressor

4. A news classifier represents each article by whether a particular word is present (1) or absent (0). Which Naive Bayes variant naturally matches this binary representation?

- A. BernoulliNB
- B. GaussianNB
- C. KNeighborsClassifier
- D. LinearRegression

5. In a training dataset, the word “prize” has never appeared in legitimate emails. Without smoothing, what problem can this cause?

- A. Its estimated likelihood under the legitimate class may be zero
- B. Every email will automatically become legitimate
- C. The number of classes will double
- D. The model will switch to KNN

6. A student sets MultinomialNB(alpha=1.0). What is the purpose of alpha in this situation?

- A. Control additive smoothing
- B. Set the number of neighboring emails
- C. Set the maximum depth of a tree
- D. Set the probability decision threshold

7. Two words, “chest” and “pain,” often appear together in clinical notes. What does the “naive” assumption simplify?

- A. It treats feature evidence as conditionally independent given a class
- B. It assumes all patients have the same diagnosis

- C. It assumes each note contains one word
- D. It removes all related words automatically

8. A dataset contains 90% legitimate emails and 10% spam. Before examining the words in a new email, which information can Naive Bayes use?

- A. The class prior probabilities
- B. The email's true label
- C. Its nearest neighbor's distance
- D. The depth of a decision tree

9. A student fits CountVectorizer on *all* emails before splitting into training and test sets. Why should the student revise this workflow?

- A. The vocabulary preparation can use information from the test emails
- B. Naive Bayes cannot work with word counts
- C. The model requires every email to contain the same words
- D. The training set must contain no text

10. A spam model achieves 96% accuracy, but nearly all spam messages are missed. Which result should the team inspect closely?

- A. Recall for the spam class
- B. The number of characters in the Python file
- C. The number of feature names alone
- D. The model's training time alone

3. Decision Trees

1. A doctor wants a model that can produce rules such as “if glucose exceeds a threshold, then check age.” Which model naturally produces this style of decision process?

- A. Decision tree
- B. KNN
- C. Naive Bayes
- D. Linear regression

2. A bank trains a decision tree to classify loan applications. The first question in the tree is whether income exceeds ₹50,000 per month. What is this first question called?

- A. A root-node split
- B. A class probability
- C. A nearest neighbor
- D. A regression coefficient

3. A student's tree correctly classifies almost every training example but performs poorly on new examples. The tree has many tiny leaves. What should the student consider?

- A. Restricting tree depth or increasing min_samples_leaf
- B. Increasing tree depth without checking validation results
- C. Removing the test set
- D. Changing all target labels to the majority class

4. A hospital wants to predict whether a patient will be readmitted: yes or no. Which Python model matches this target?

- A. DecisionTreeClassifier
- B. DecisionTreeRegressor
- C. KNeighborsRegressor
- D. LinearRegression

5. The same hospital now wants to predict the *number of days* a patient will remain admitted. Which tree model matches this target?

- A. DecisionTreeRegressor
- B. DecisionTreeClassifier
- C. BernoulliNB
- D. KNeighborsClassifier

6. A tree uses the rule temperature ≤ 38.0 . A patient has a temperature of 37.5°C. What does the tree do?

- A. Follow the branch where the condition is true
- B. Follow both branches
- C. Stop making predictions
- D. Change the temperature to 38.0°C

7. A team needs to explain a small tree's predictions to clinicians. Which Python function can help display its learned splits?

- A. plot_tree
- B. train_test_split
- C. StandardScaler
- D. accuracy_score

8. A student sees that a feature has high feature_importances_ in a fitted tree. Which conclusion is justified?

- A. The fitted tree relied strongly on that feature for its splits
- B. The feature has been proven to cause the outcome
- C. The feature will always be important in every dataset
- D. The feature should be removed immediately

9. A tree for predicting disease at hospital admission includes a laboratory result obtained three days after admission. What is the main concern?

- A. The feature would not be available when the prediction is needed
- B. The laboratory result is numerical
- C. A tree cannot use medical data
- D. The tree has too few classes

10. A student wants to compare max_depth values of 2, 4, and 8. What is the best purpose of doing so?

- A. Find a depth that balances useful patterns and generalization
- B. Force all trees to make identical predictions
- C. Determine the number of target classes
- D. Avoid fitting the model

4. Random Forest

1. A single decision tree gives noticeably different predictions when the training data changes slightly. A team combines many trees trained with randomness. What model are they building?

- A. Random forest
- B. Logistic regression
- C. Gaussian Naive Bayes
- D. KNN regression

2. A researcher creates `RandomForestClassifier(n_estimators=100)`. What does `n_estimators=100` specify?

- A. The forest contains 100 trees
- B. The training set has 100 rows
- C. The target has 100 classes
- D. Each tree has depth 100

3. A forest classifies a transaction as fraudulent when most of its trees vote “fraud.” What is this an example of?

- A. Combining predictions from multiple trees
- B. Finding the nearest transaction
- C. Using a single linear equation
- D. Removing the minority class

4. A property company wants to predict a house’s numerical sale price using many trees. Which class should it use?

- A. `RandomForestRegressor`
- B. `RandomForestClassifier`
- C. `BernoulliNB`
- D. `KNeighborsClassifier`

5. A student asks why each tree in a forest does not receive exactly the same training rows. What is the purpose of bootstrap sampling?

- A. To create variation by sampling rows with replacement
- B. To copy the test set into training
- C. To make every tree identical
- D. To standardize every feature

6. A fraud-detection dataset has 99% legitimate transactions. A forest gets 99% accuracy but detects almost no fraud. What should the team examine?

- A. Precision and recall for the fraud class
- B. Only the total number of trees
- C. Only the accuracy score
- D. Only the names of the features

7. A fitted forest reports high importance for “customer age.” What is a careful interpretation?

- A. The forest used age substantially, but the result does not prove age causes the outcome
- B. Age has been proven to cause the outcome
- C. Every future dataset will give age the same importance
- D. Age is necessarily the only feature needed

8. A researcher sets `oob_score=True` while using bootstrap sampling. Which samples contribute to the out-of-bag estimate for a particular tree?

- A. Training samples not selected for that tree's bootstrap sample
- B. The final test samples
- C. Only samples with missing labels
- D. Every sample used to fit that tree

9. A team increases `n_estimators` from 50 to 500. Which trade-off should they expect?

- A. More computation, without a guarantee of better test performance
- B. A guaranteed increase in test accuracy
- C. Automatic conversion into a single tree
- D. Removal of all class imbalance

10. A hospital forest uses a feature recorded after a patient is discharged to predict risk at admission. The validation score is excellent. What should the team do?

- A. Remove the future-information feature and reevaluate
- B. Deploy immediately because the score is high
- C. Add more copies of that feature
- D. Increase the number of trees until the problem disappears

5. Logistic Regression

1. A clinic wants to predict whether a patient will develop a condition within one year: yes or no. Which model is a suitable simple classification baseline?
 - A. LogisticRegression
 - B. LinearRegression for an unrestricted numerical output
 - C. RandomForestRegressor
 - D. KNeighborsRegressor
2. A fitted logistic regression model gives a patient a positive-class probability of 0.78. The clinic uses a threshold of 0.5. What class will the system assign?
 - A. Positive
 - B. Negative
 - C. Both classes
 - D. No class
3. A screening program is especially concerned about missing patients who have a disease. What might it consider doing to its positive decision threshold?
 - A. Lowering it, while evaluating the resulting false alarms
 - B. Raising it to 1.0 without evaluation
 - C. Removing the threshold entirely
 - D. Setting every feature to zero
4. A student lowers the positive-class threshold from 0.5 to 0.2. What is a possible result?
 - A. More actual positive cases are identified, along with more false alarms
 - B. All false positives disappear
 - C. The model becomes a decision tree
 - D. The fitted coefficients are automatically retrained
5. A model uses age in years and income in rupees. The student wants regularization and optimization to behave sensibly across features with very different scales. What should they consider?
 - A. Standardizing the input features
 - B. Removing the target labels
 - C. Setting `n_estimators=100`
 - D. Choosing an odd `K`
6. A student writes `LogisticRegression(C=0.01)` and compares it with `LogisticRegression(C=100)`. Which setting generally applies stronger regularization?
 - A. `C=0.01`
 - B. `C=100`
 - C. Both always apply identical regularization
 - D. Neither model uses regularization
7. A fitted model has a positive coefficient for blood pressure when predicting a condition. What does that coefficient suggest, with the other supplied features held fixed?
 - A. Higher blood pressure increases the model's log-odds of the positive class
 - B. Blood pressure is proven to cause the condition
 - C. Higher blood pressure must reduce the predicted probability
 - D. The model ignores blood pressure

8. A disease occurs in only 1% of patients. A model predicts “no disease” for everyone and achieves 99% accuracy. What should the student check?

- A. Recall, precision, and the confusion matrix for the disease class
- B. Only overall accuracy
- C. The number of Python imports
- D. Only the number of features

9. While training, scikit-learn reports that logistic regression did not converge. What is a sensible next step?

- A. Check feature scaling and consider solver settings or `max_iter`
- B. Assume the model is perfectly trained
- C. Set `n_neighbors=5`
- D. Increase the number of trees

10. A hospital model predicts disease at admission but uses the final discharge diagnosis as an input. Why is its impressive test result questionable?

- A. The input contains information unavailable at admission
- B. Logistic regression cannot use categorical targets
- C. The model needs at least 100 features
- D. Predicted probabilities can never be evaluated